SOLUTION QUIZ-01 BSE-6A

QUESTION-01

1. C
2. B
3. B
4. C
5. A

QUESTION-02

a. Transmission Delay = Number of Cars x Service Time per Car = 20 cars x 2 sec/car = 40s

b. Propagation Delay = Distance / Speed = 400 km / 10 kms^{-1} = 40s

c.

   1. **Car 1 leaves Booth 1:** At t = 2 s.
   2. **Requirement for Booth 2:** Service cannot start until the entire caravan (all 20 cars) has arrived at Booth 2.
   3. **When does the last car (Car 20) arrive at Booth 2?**
   4. Car 20 leaves Booth 1 at t = 40s (calculated in part a).
   5. Car 20 travels for 40s (calculated in part b).
   6. Car 20 arrives at Booth 2 at t = 40 + 40 = 80s.
   7. **Service Start:** Booth 2 begins service at t = 80s.

   Calculation:
   Time from Car 1 leaving Booth 1 (t=2) to Car 1 entering service at Booth 2 (t=80).
   Time = 80s - 2s = 78s

d. No.

   - Booth 1 is busy servicing cars from t = 0 to t = 40.
   - Booth 2 cannot start servicing cars until the entire caravan arrives (at t=80). So Booth 2 works from t = 80 to t = 120.

e. Yes. During the 40-second gap, the entire caravan is effectively "on the road" (propagating) or waiting in the queue at Booth 2, but no tollbooth is actively servicing a car.